

Vinicius Francisco da Silva

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Master's degree in Computer Science from the PPGCC (Graduate Program in Computer Science) at UFMG (Federal University of Minas Gerais), where I conducted research at LaC (Compilers Lab) under the supervision of Dr. Fernando Quintão Pereira. Currently pursuing a postgraduate specialization in Data Science at UNICAMP (State University of Campinas).

AREAS OF INTEREST

- Compilers
- Code Optimization
- Formal Verification
- Virtualization and Containerization
- Operating Systems
- Programming Language Theory

EDUCATION

Federal University of Minas Gerais (UFMG) <i>MSc Student</i>	Belo Horizonte, Minas Gerais <i>9/2024 - 08/2026</i>
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- **Research Title:** Multi-Language Benchmark Generation via L-Systems
- **Supervisor:** Fernando Magno Quintão Pereira

Postgraduate Lato Sensu (UNICAMP) <i>Data Science</i>	Remote <i>8/2024</i>
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Federal University of Minas Gerais (UFMG) <i>Individual Course Enrollment</i>	Belo Horizonte, Minas Gerais <i>Completed</i>
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- **Subject:** Static Program Analysis

Pontifical Catholic University of Minas Gerais (PUCMINAS) <i>Bachelor's Degree in Computer Science</i>	Belo Horizonte, Minas Gerais <i>Completed</i>
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- **Undergraduate Final Project Title:** Maximizing Efficiency in Inter-Process Communication: Exploring a Zero-Copy Abstraction
- **Supervisor:** Pedro Henrique Penna

EXPERIENCE

Researcher at Compilers Lab (LaC) - DCC/UFMG 9/2026 – Current

- **Research topic:** Searching for GPU Thread-Block and Tile-Size Configurations in PolyBench-MLIR via Coordinate Descent

Graduate Research at Compilers Lab (LaC) - DCC/UFMG 8/2024 – 08/2026

- **Research topic:** Multi-Language Benchmark Generation via L-Systems

Jr. Linux Kernel Developer 6/2024 – 10/2024
MagaluCloud *Remote*

Mid-Level FullStack Developer 3/2021 – 6/2024
Sociedade Mineira de Cultura *Belo Horizonte, Minas Gerais*

Intern Researcher Compilers and Operating System 06/2020 - 07/2021
Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS) *Belo Horizonte, Minas Gerais*

PIBIC/CNPQ Researcher at Graduate Program in Dentistry 10/2019 - 8/2020
Institute of Biology Sciences - (ICBS) - (PUCMINAS) *Belo Horizonte, Minas Gerais*

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Tutor - Compilers discipline 4/2021 - 6/2021
Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS) *Belo Horizonte, Minas Gerais*

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Tutor - Algorithms and Data Structures II discipline 10/2021 - 01/2022
Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS) *Belo Horizonte, Minas Gerais*

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Tutor - Data Base discipline 7/2020 - 11/2020
Institute of Exact Sciences and Informatics - (ICEI) - (PUCMINAS) *Belo Horizonte, Minas Gerais*

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VOLUNTEERING

Artifact Evaluator <i>International Symposium on Code Generation and Optimization (CGO27)</i>	8/2s26 - Current <i>Remote</i>
Artifact Evaluator <i>International Conference on Parallel Architecture and Compilation Techniques (PACT25)</i>	8/2025 - 9/2025 <i>Remote</i>
Artifact Evaluator <i>International Symposium on Code Generation and Optimization (CGO26)</i>	9/2025 - 12/2025 <i>Remote</i>
Artifact Evaluator <i>International Conference on Compiler Construction (CC26)</i>	12/2025 - 12/2025 <i>Remote</i>
Artifact Evaluator <i>Programming Language Design and Implementation (PLDI26)</i>	1/2026 - 04/2026 <i>Remote</i>

PUBLICATIONS

V. Silva, Soares, and Fernando Magno Quintão Pereira 2026 paper
Vinícius Francisco da Silva, Leite, and Pereira 2025 paper
V. F. da Silva 2024 paper

REFERENCES

- Silva, Vinícius et al. (2026). *Memory Allocation for Constant-Bounded Programs*. arXiv: 2608.14471 [cs.PL]. URL: <https://arxiv.org/abs/2608.14471>.
- Silva, Vinícius Francisco da et al. (2025). “Multi-Language Benchmark Generation via L-Systems”. In: *arXiv preprint arXiv:2512.17616*.
- Silva, V. F. da (2024). *Maximizando a Eficiência na Comunicação entre Processos: Explorando uma Abstração de Zero-Cópia*. Tech. rep. Belo Horizonte, Brazil: Pontifícia Universidade Católica de Minas Gerais (PUC Minas).

HONORS & AWARDS

Featured Paper – Brazilian Symposium on Programming Language (SBLP26)

- Issued by Brazilian Computer Society (SBC)
- Paper “*Multi-Language Benchmark Generation via L-Systems*” Selected as the Second Best Paper at the Brazilian Symposium on Programming Language (SBLP26).
- [view](#)

LETTER OF RECOMMENDATION

Pedro Henrique Penna

- [view](#)

Huimin Cui and Gabriel Rodríguez

- [view](#)

Olivia Hsu

- [view](#)

Nanvix Operating System Internship

PACT26 Artifact Evaluation Volunteership

CGO26 Artifact Evaluation Volunteership

SOFTWARE PATENTS

Software Patent - BR512022000810-6

- [view](#)

AnalisedoSorriso Mobile App

Software Patent - BR512022000377-5

- [view](#)

3DInteractiveDentalAnatomy Mobile App

Software Patent - BR512021000181-8

- [view](#)

InfoOdonto Mobile App

Software Patent - BR512020002866-7

- [view](#)

DTMOne Mobile App

Software Patent - BR512019000555-4

- [view](#)

OrthoHelp Mobile App

CONTRIBUTIONS

V Programming Language

Contributor

Open Source

GitHub

- **Merged PRs:** [View on GitHub](#)

Carbon Programming Language

Contributor

Open Source

GitHub

- **PRs:** [View on GitHub](#)

LLVM

Contributor

Open Source

GitHub

- **PRs:** [View on GitHub](#)

LLVM/CIRCT

Contributor

Open Source

GitHub

- **PRs:** [View on GitHub](#)

LLVM/EUDSL

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